

Excercise - 04

Setting up a Local Area Network (LAN)

Aim -

To design, set up, and configure a Local Area Network (LAN) connecting four computers through a network switch, assign static IP address to all devices, verify connectivity using the ping command and demonstrate resource/data sharing across the network.

Equipment Required:

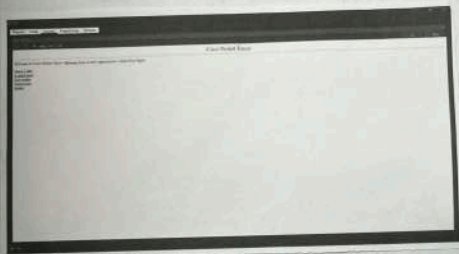
S.No	Equipment	Quantity	Specification
1	Personal Computer (PC)	4	Ethernet (NIC) enabled
2	Network Switch	1	8/16/24 port, unmanaged/managed
3	Ethernet cable (Cat 5e Cat 6)	4	Straight-through, RJ-45 connectors

Theory:

A Local Area Network (LAN) is a Network that connects computers and devices within a limited geographical area such as home, office, or laboratory, allowing them to share resources such as files, printers and Internet connections.

Network Diagram :

The LAN was implemented using a star topology as shown below, with 4 PCs connected individually to a central switch.



Implementation in Cisco packet Tracer

Since physical hardware was limited, the LAN was also implemented and verified using Cisco packet Tracer simulation software, following the steps below.

Step 1: place the Devices

1. Open Cisco packet Tracer
2. From the devices category panel, select End Device and drag 4 PCs onto the workspace.
3. Select Switches and drag 1 switch onto the workspace.

Step 2: cable the devices

4. click connection $\rightarrow$ Select copper Straight-through cable.
5. click each PC, then click the switch, and select an available Fast Ethernet port of connection each computer.
6. Repeat for all four PCs. Green link lights confirm a successful physical connection.

Step 3: Assign IP addresses to PCs

7. click a PC $\rightarrow$ Desktop tab $\rightarrow$ IP configuration
8. Enter the static IP address and Subnet Mask per the IP table (section 5) for each PC.

Step 4: configure the switch Management IP

9. click the switch $\rightarrow$ CLI Tab
10. Enter the following commands to assign a management IP

enable

configure terminal

interface Vlan 1

ip address 10.1.1.5 255.0.0.0

no shutdown

exit

Step 5: Verify connectivity using ping

11. click a pc $\rightarrow$ Desktop tab $\rightarrow$ command prompt
12. use the ping command to test connectivity to other PCs
13. Successful "Reply from ..." message confirms connectivity.

Step 6: Demonstrate Data Sharing (Server Simulation)

14. Add a server device (End Device $\rightarrow$ server-Pt) and connect Pt to the switch.
15. Assign Pt an IP address (eg, 10.1.1.10, subnet mask 255.0.0.0) via Pt Desktop $\rightarrow$ IP configuration tab.
16. Open the ~~switch~~ (config) tab and enable the HTTP services
17. From another PC, open the web browser (desktop tab) and

SLK